

Magneto-dynamics

Recall, $\vec{j} = \rho \vec{v}_d$

The density of electrons * the drift velocity.

We can look at a wire with current in the rest frame of the nuclei, in which case the electrons are flowing in the direction opposing current. A single electron outside the wire would experience a force

$$\vec{F}_b = q\vec{v} \times \vec{B} \quad (1)$$

$$\vec{B} = \frac{\mu_0 i}{2\pi r} \quad (2)$$

$$i = ja = \rho v A \quad (3)$$

$$F_b = \frac{e\mu_0 \rho v^2 A}{2\pi r} (-\hat{y}) \quad (4)$$

If we change our frame to have the electrons stationary, so there is no magnetic force.

$$\vec{F}_b = -e\vec{v} \times \vec{B} \quad (5)$$

$$= 0 \quad (6)$$

If we don't have a magnetic field, we are forced to conclude that there is an electric field, which is causing a force on the electron equal to what we see in the other rest frame.

Remember back to special relativity. In the frame where the electrons are moving, the length is L . In the frame where electrons are stationary, the distance between electrons is γL , whereas the distance between nuclei is $\frac{L}{\gamma}$.

We can now look at the charge densities.

$$\rho_+ = \rho_- \quad (7)$$

$$\rho'_+ > \rho'_- \quad (8)$$

$$\rho'_+ = \frac{Q}{\frac{AL}{\gamma}} = \gamma \rho_+ = \gamma \rho_- \quad (9)$$

$$\rho'_- = \frac{Q}{A\gamma L} = \frac{1}{\gamma} \rho_- \quad (10)$$

$$(11)$$

Which gives us the result

$$\rho'_{tot} = \rho'_+ - \rho'_- \quad (12)$$

$$= \gamma \rho_- - \frac{1}{\gamma} \rho_- \quad (13)$$

$$\rho'_{tot} = \frac{\gamma v^2}{c^2} \rho_- > 0 \quad (14)$$

In the rest frame of the electrons, we can use gauss's law.

$$\oiint \vec{E}' \cdot d\vec{A}' = \frac{Q'_{enc}}{\epsilon_0} \quad (15)$$

We can say by symmetry that the electric field can only depend on r and point in the $\hat{r}$ direction. We can draw a cylinder around the wire.

$$E'(r')(2\pi r' l') = \frac{\rho_{tot} l' A}{\epsilon_0} \quad (16)$$

$$r' = r \quad (17)$$

$$E'(r) = \rho'_{tot} \frac{A}{2\pi r \epsilon_0} \quad (18)$$

$$= \frac{\gamma v^2 \rho_- A}{2\pi r \epsilon_0 c^2} \quad (19)$$

So the electric force in the primed frame is

$$\vec{F}'_e = \frac{\gamma e v^2 \rho_- A}{2\pi r \epsilon_0 c^2} (-\hat{y}) \quad (20)$$

We have to transform the forces to be in the same frame so that we can compare them.

$$F_y = \frac{1}{\gamma} F_{y'} \quad (21)$$

so we get

$$\vec{F}_e \stackrel{?}{=} \vec{F}_b \quad (22)$$

$$\frac{ev^2\rho_A}{2\pi r\epsilon_0 c^2}(-\hat{y}) \stackrel{?}{=} \frac{e\mu_0\rho_A v^2}{2\pi r}(-\hat{y}) \quad (23)$$

$$\mu_0 \frac{ev^2\rho_A}{2\pi r}(-\hat{y}) \stackrel{?}{=} \frac{1}{\epsilon_0 c^2} \frac{ev^2\rho_A}{2\pi r}(-\hat{y}) \quad (24)$$

$$\mu_0 = \frac{1}{\epsilon_0 c^2} \quad (25)$$

$$\text{plug in values, we get that this is true!} \quad (26)$$

1 Electric and magnetic fields are not two entirely separate things, but aspects of a single phenomenon - ELECTROMAGNETISM!
